7 Read the passage and answer the questions that follow.

Traditional wind turbines are limited by their height; the friction of the Earth's surface slows wind speeds at low altitudes, and the massive concrete foundations required for towers over 150 metres tall are both expensive and environmentally invasive. To overcome these limitations, engineers have developed **Airborne Wind Energy Systems (AWES)**.

In early 2026, the S2000 Stratosphere Airborne Wind Energy System (SAWES), developed in China, became the world's first megawatt-class "flying power plant" to successfully feed electricity into a national grid. The system utilizes a helium-filled, ring-shaped aerostat (a "balloon") anchored to a ground station by a high-strength tether. A simplified schematic of the S2000 system is shown in Fig. 7.1. The aerostat remains aloft through upthrust, the upward force exerted by the surrounding air on the balloon when it is immersed in the air. This allows the turbines to access the "jet effect" of the stratosphere—a region where winds are faster, more stable, and carry significantly more kinetic energy than those at ground level.

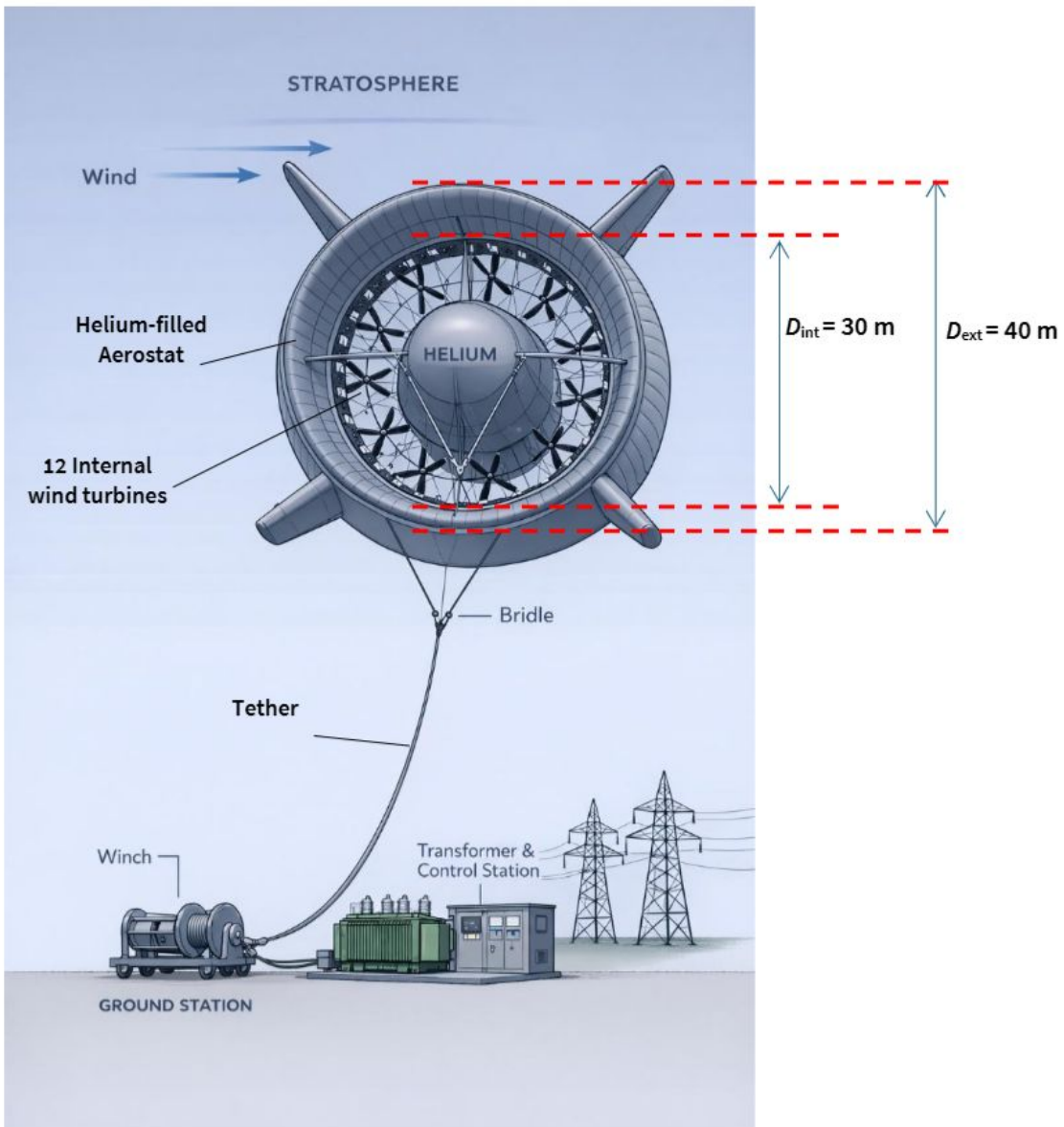


Fig. 7.1: Simplified Schematic of the S2000 System

The S2000 is linked to its ground-based winch station by a 2.5 km multipurpose conductive tether. This cable is a composite structure, featuring a high-tensile carbon-fibre core to withstand buoyancy and wind-drag forces, surrounding an aluminium-based conductive layer for power transmission. Onboard sensors relay real-time data through fibre-optic filaments within the tether, allowing a ground-based AI to adjust turbine speeds and aerostat pitch.

Table 7.1: S2000 SAWES System Specifications and Operating Parameters

Category	Parameter	Value (at h = 2000 m)
Atmospheric	Air density (ρ_{air})	1.007 kg m^{-3}
	Average Wind Speed (v)	15.0 m s^{-1}
Aerostat	External Diameter (D_{ext})	40 m
	Internal Diameter (D_{int})	30 m
	Volume of Helium (V_{He})	20,000 m^3
	Aerostat Structural Mass ($m_{structure}$)	5,000 kg
Generation	Total Turbine Swept Area ($A_{turbine}$)*	340 m^2
	Turbine Assembly Mass ($m_{turbine}$)*	2,400 kg
	Maximum Rated Power (P_{max})	3.0 MW
	Average Power Output from Initial Test	770 kW (over 30 min)
Tether	Total Length l_{tether}	2500 m
	Linear Mass	1.8 kg m^{-1}

*Note: Total of the 12 internal turbines.

Additional Constants and Parameters:

Aluminium resistivity: $\rho_{Al} = 2.8 \times 10^{-8}\ \Omega$ m

Helium density: $\rho_{He} =$ 0.18 kg m^{-3}

Power coefficient: $C_p = 0.40$ (maximum)

System efficiency: $\eta_{system} = 45\%$ (maximum)

Drag coefficient (ring aerostat): $C_d = 0.25$

Energy density of the stratospheric wind, which represents the kinetic energy per unit volume of moving air, helps engineers assess the energy resource available at different altitudes and locations. This, in turn, will help to determine the amount of power that can be harnessed from the wind through the turbine.

Real turbines cannot extract all available wind power. The power coefficient (C_p) represents the fraction of wind power actually converted to mechanical power. For example, a C_p of 0.40 means the turbine converts 40% of the wind's kinetic energy into mechanical rotation. System efficiency (η_{system}) accounts for subsequent losses during electrical conversion and transmission, reducing the final delivered power. This includes losses in the generator, power electronics, and the resistance of the 2.5 km tether. A system efficiency of 45% means only 45% of the mechanical power extracted by the turbines reaches the ground station as usable electricity.

(a)	**(i)**	Using data from Table 7.1, calculate the buoyant force (upthrust) $\rho_{air} V_{He} g$ acting on the helium-filled aerostat at its operating altitude.
		$\rho_{air} V_{He} g = 1.007 \times 20{,}000 \times 9.81 = 197.57 \times 10^3$ N [A1]
		buoyant force = ... N [1]
	(ii)	Calculate the total weight of Helium-filled aerostat.
		Weight of helium = 0.18 × 20,000 × 9.81 = 35,316 N Weight of aerostat = 5,000 × 9.81 = 49,050 N Weight of turbines = 2,400 × 9.81 = 23,544 N Total weight = 107,910 N [A1] = 108, 000 N
		total weight = ... N [2]
	(iii)	Using your answers from **(a)(i)** and **(a)(ii)**, determine the minimum tension in the tether.
		minimum tension = 197,570 − 107,910 = 89,660 N = 90,000 N [A1]
		minimum tension = ... N [1]

(iv) Calculate the weight of the tether.

weight of tether = 2,500 × 1.8 × 9.81 = 44,145 = 44,000 N

weight = ... N [1]

(v) State whether the force exerted by the Winch on the tether will be greater, equal or smaller than your answer in **(a)(iii)**, explain your answer.

. The force exerted by the Winch on the tether will be smaller. [A1]

. Since the tether is in equilibrium, (considering the forces on the tether), the force on the tether upwards by the aerostat [answer in (a)(iii)] is equal to the weight of the tether and the force exerted by the Winch on the tether (downwards) [M1]

.

..... [2]

(b) **(i)** When wind flows around the ring-shaped aerostat, it experiences a drag force given by $\frac{1}{2}\rho v^2 C_d A$, where A is the frontal area presented to the wind. Assuming the frontal area is a circle with external diameter D_{ext}, calculate the drag force acting on the aerostat in the stratospheric wind conditions.

Frontal area = $\frac{\pi D_{ext}^2}{4} = \frac{\pi(40^2)}{4} = 1256.6\ \text{m}^2$ [A1]

Drag force, $F_D = \frac{1}{2}\rho v^2 C_d A = \frac{1}{2}(1.007)(15.0)^2(0.25)(1256.6) = 35590\ \text{N} \approx 36\ \text{kN}$ [A1]

[Both the frontal area calculation and the substitution into the drag force formula must be shown explicitly. Award zero if working is absent even if the answer is correct.]

drag force = ... N [2]

(ii) Determine the magnitude of the change in tension in the tether, when drag force acts on the aerostat. Explain your answer.

Magnitude of the change in tension in the tether = 36 kN [A1]

Tension is a vector, a change in tension $\Delta \boldsymbol{T}$ will be vector subtraction given by $(\boldsymbol{T}_f - \boldsymbol{T}_i)$, which will be the opposite of the drag force. Therefore, magnitude of the change in tension $|\Delta \boldsymbol{T}|$, is 36 kN. [B1]

Alternatively, since $\boldsymbol{T}_f = \boldsymbol{T}_i - \boldsymbol{D}$, we have $\boldsymbol{T}_f - \boldsymbol{T}_i = -\boldsymbol{D}$, which implies that the drag force D causes the change in tension in the tether.

magnitude of change in tension = ... N [2]

(c) (i) By considering the energy density of the stratospheric wind, show that the power that can be harnessed from the wind can be expressed as $\frac{1}{2}\rho_{air}A_{turbine}v^3$, where ρ_{air}, $A_{turbine}$ and v are quantities expressed in Table 7.1.

energy density $= \left(\frac{1}{2}mv^2\right)_{air} / V_{air} = \frac{1}{2}\rho_{air}v^2$ [A1]

$P = (\text{energy density})(\text{air volume change rate}) = \left(\frac{1}{2}\rho_{air}v^2\right)(A_{turbine}v) = \frac{1}{2}\rho_{air}A_{turbine}v^3$

[2]

(ii) Determine the total power harnessed from the wind through the turbine.

$P = \frac{1}{2}\rho_{air}A_{turbine}v^3 = \frac{1}{2}(1.007)(340)(15.0)^3$

$= 577.8 \times 10^3$ W [A1]

total power = ... W [1]

(iii) Hence or otherwise, determine maximum electrical power delivered to ground.

Mechanical power = $C_p \times P = 0.40 \times 577.8 \times 10^3 = 231.1 \times 10^3$ W [B1]

Delivered power = $\eta_{system} \times 231.1 \times 10^3 = 0.45 \times 231.1 \times 10^3 = 104.0 \times 10^3$ W [A1]

power = ... W [2]

(iv) The average power output from the Initial test from Table 7.1 is 770 kW. State whether this is higher or lower than your calculated value in **(c)(iii)** and suggest a reason for the difference.

Initial test shows 770 kW average (from Table 7.1), much higher than calculated value. [B1]

Difference suggests variable wind conditions with periods of higher wind speeds during 30 min testing period. [B1] $\left\{P = (\text{energy density})(A_{turbine}v) = \frac{1}{2}\rho_{air}A_{turbine}v^3\right\}$

..... [2]

(d) Dimensions of the tether must be considered in order to carry a current of up to 300 A and heat losses through the tether only account for 0.48% of the maximum rated power (P_{max}) for efficient transmission of power.

Determine the cross-sectional area of the aluminium needed in the conductive layer of the tether that satisfies the above considerations.

$$\frac{0.48}{100} = \frac{P_{loss}}{P_{max}} \Rightarrow P_{loss} = \frac{0.48}{100} \times \left(3.0 \times 10^6\right) = 14.4 \text{ kW}$$

$$P_{loss} = I^2 R \Rightarrow R = \frac{P_{loss}}{I^2} = \frac{14400}{300^2} = 0.16\ \Omega$$

$$R = \rho_{Al} \frac{l_{tether}}{A_{tether}}$$

$$\Rightarrow 0.16 = 2.8 \times 10^{-8} \left(\frac{2500}{A_{tether}}\right)$$

$$\therefore A_{tether} = 4.4 \times 10^{-4}\ \text{m}^2$$

cross-sectional area = ... m^2 [2]

[Total: 20]